Cryptography and Network Security (BT14503) :

Modern Symmetric-Key Ciphers

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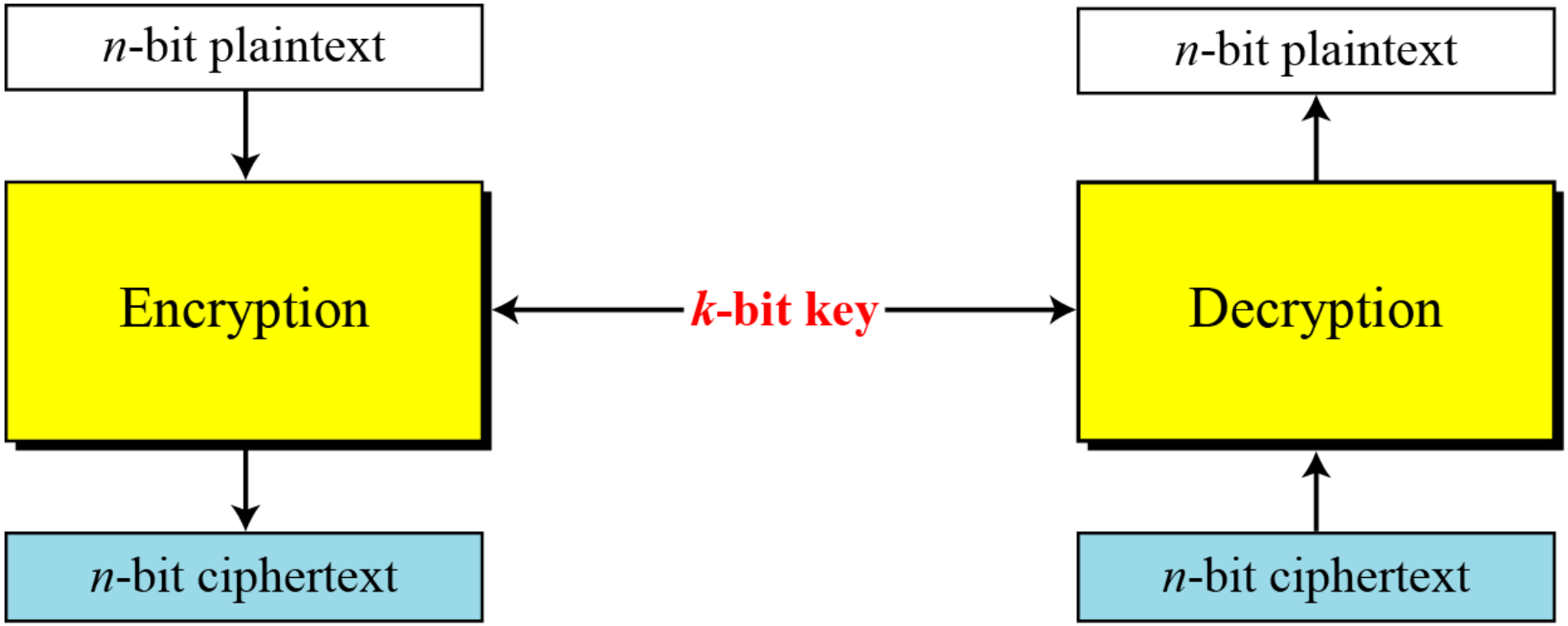
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Modern Block ciphers

A symmetric-key modern block cipher encrypts an n-bit block of plaintext or decrypts an n-bit block of ciphertext. The encryption or decryption algorithm uses a k-bit key.

Modern Block ciphers…



Modern Block ciphers…

- If the message has fewer than n bits, padding must be added to make it an n-bit block
- if the message has more than n bits, it should be divided into n-bit blocks
  - and the appropriate padding must be added to the last block if necessary.
- The common values for n are 64, 128, 256, or 512 bits.

Substitution or Transposition

- A modern block cipher can be designed to act as a substitution cipher or a transposition cipher.
- Which one do you think is better? Why?

Substitution or Transposition

To be resistant to exhaustive-search attack, a modern block cipher needs to be designed as a substitution cipher.

Modern Block Ciphers..

- Full-size Key Ciphers
  - Full-size Key Transposition Block Ciphers
  - Full-size Key Substitution Block Ciphers
- Partial-size Key Ciphers
- Keyless Ciphers
  - Keyless Transposition Ciphers
  - Keyless Substitution Ciphers

Modern Block Ciphers..

- **Full-Size Key Transposition Block Ciphers**
  - Transposes bits without changing their values
    - modelled as an n-object permutation with $n!$ permutation tables.
  - Need to have n! possible keys,
    - the key should have $\lceil \log_2(n!) \rceil$ bits.

Modern Block Ciphers..

- **Full-Size Key Transposition Block Ciphers Example:**
  - Permutation tables for a 3-bit block transposition cipher where the block size is 3 bits.
    - The set of permutation tables has 3! = 6 elements
    - Key should have $\lceil \log_2(n!) \rceil$ bits = $\lceil \log_2(3!) \rceil$ bits = 3 bits.

Modern Block Ciphers..

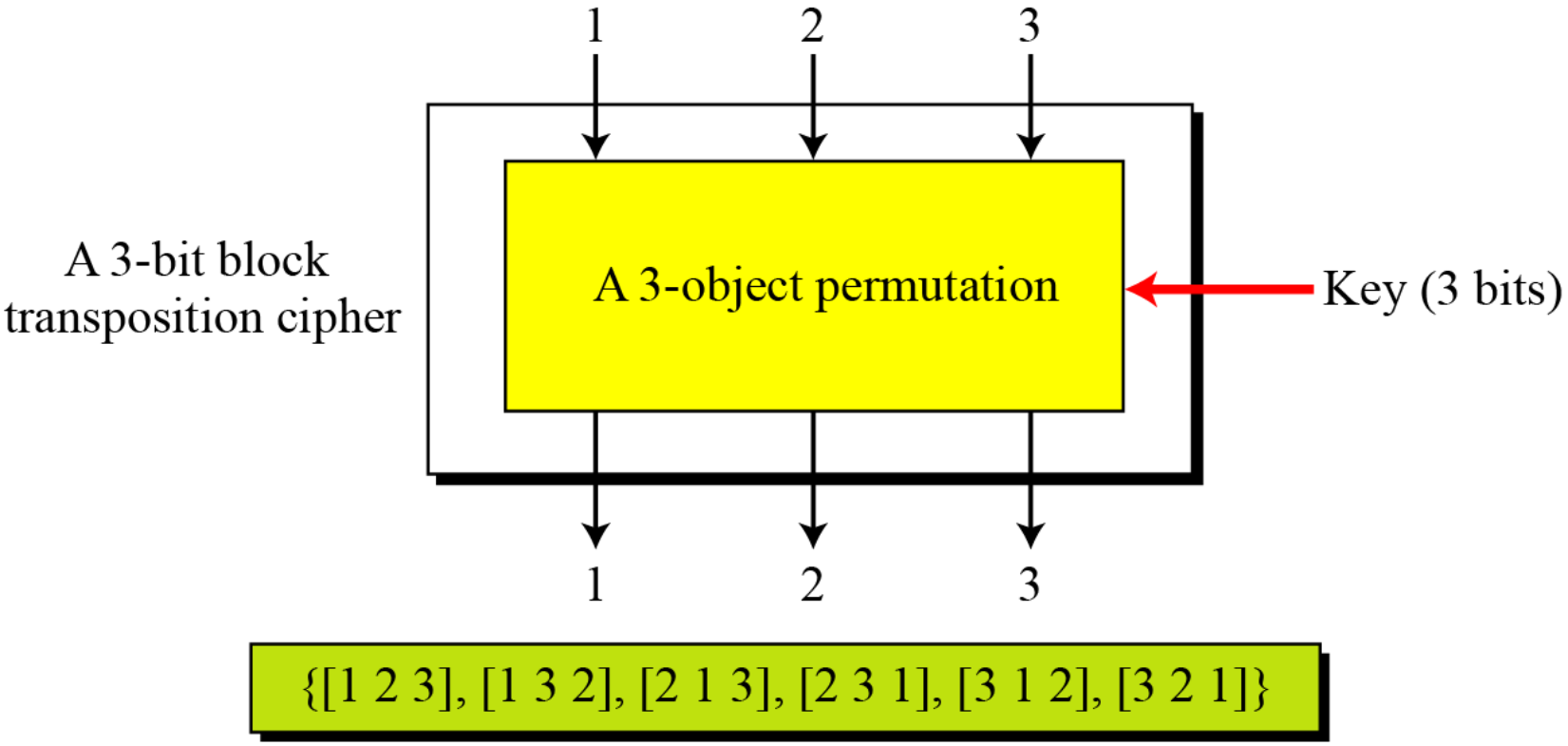


Fig 1: Transposition block cipher modeled as permutation

Modern Block Ciphers..

- **Full-Size Key Substitution Block Ciphers**
  - A full-size key substitution cipher does not transpose bits; it substitutes bits.
  - Model substitution cipher as permutation using encoding and decoding
  - Decoding
    - transforms n-bit integer into 2^n-bit string with single 1 and $2^n - 1$ 0's.
    - This position of single 1 is the value of integer, in which position range from 0 to $2^n - 1$
  - Encoding is reverse of decoding
  - Cipher is modelled as permutation of $2^n!$ objects

Modern Block Ciphers..

- A full-size key n-bit transposition cipher or a substitution block cipher can be modeled as a permutation, but their key sizes are different:
  - Transposition: the key is $\lceil \log_2(n!) \rceil$ bits
  - Substitution: the key is $\lceil \log_2(2^n!) \rceil$ bits

Block Ciphers as Permutation Groups

- It should be noted that full-size key transposition or substitution cipher forms permutation group
  - meaning, if encryption and decryption uses more than one stage of any of these ciphers, the result is equivalent to permutation group under the composition operation.
- Two or more cascaded permutations can always be replaced by single permutation.
  - This is because permutation forms a group under the composition operation.
- Multiple applications of this ciphers has the same effect as a single application of the transformation
- This means that it is useless to apply more than one stage of full-size key cipher,
  - Because the effect is same as having a single stage.

Block Ciphers as Permutation Groups...

- **Partial-size key ciphers**
  - Actual ciphers can not use full-size keys because the size of the key becomes large
    - especially substitution cipher
- **Partial-size key ciphers** is group under the composition operation
  - if it is a subgroup of the corresponding full-size cipher

Block Ciphers as Permutation Groups...

- **Keyless Cipher:**
  - Keyless ciphers are used as components of keyed ciphers
  - Considered as building blocks of modern block ciphers
  - These are of two types:
    - **Keyless Transposition Cipher:**
      - Also called as P-boxes
      - uses single permutation rule (can be represented by table)
    - **Keyless Substitution Cipher**
      - Also called as S-boxes
      - uses mathematical function (rep. by table) to map input to output

Components of a Modern Block Cipher

- Modern block ciphers normally are keyed substitution ciphers in which the key allows only partial mappings from the possible inputs to the possible outputs.
- P-Boxes
  - A P-box (permutation box) parallels the traditional transposition cipher for characters.
  - It transposes bits.

Components of a Modern Block Cipher…

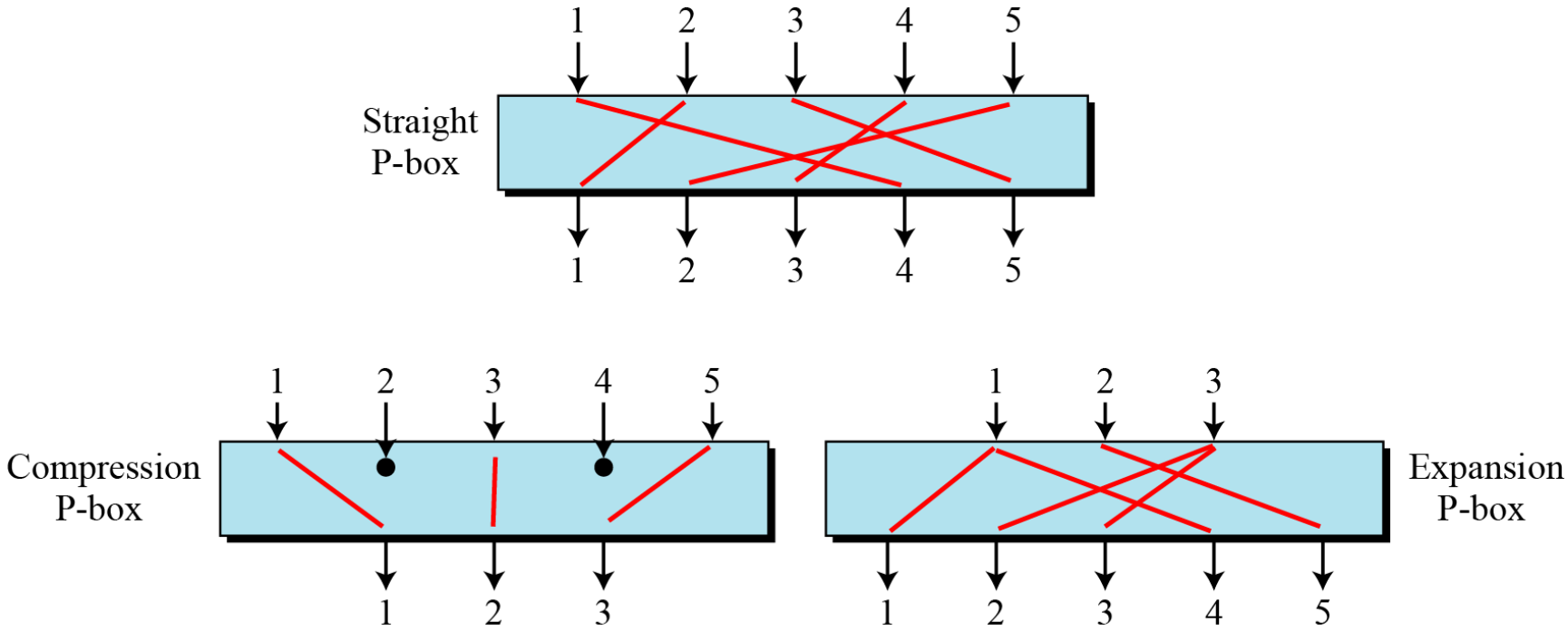


Fig 1: Three Types of P-boxes

P-Boxes..

- Possible mappings of a straight P-Box

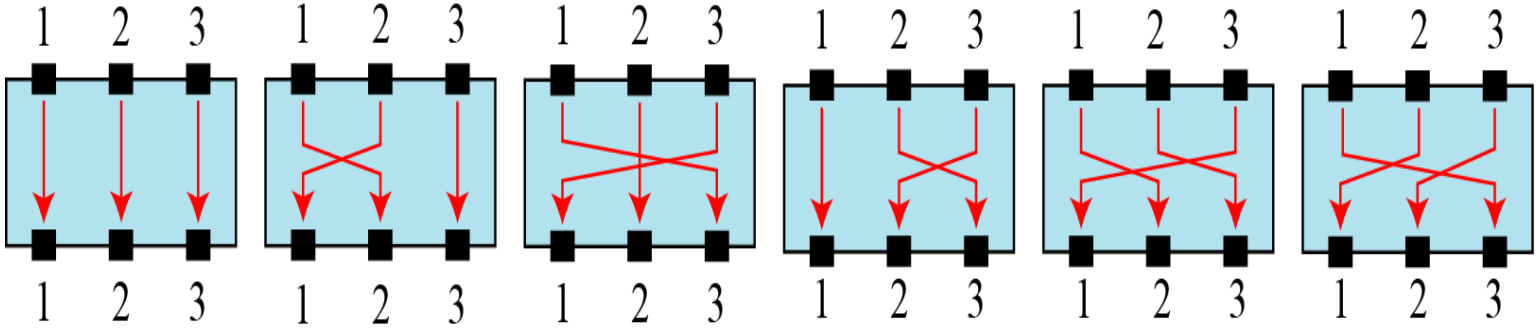


Fig 2: The possible mappings of 3 X 3 P-box

P-Boxes..

- Straight P-Box
  - n inputs and generates n outputs as permutation
  - Possible $n!$ mappings
  - These are normally keyless
  - If implemented in hardware, it is prewired
  - If implemented in software, permutation table provides rule of mapping

58	50	42	34	26	18	10	02	60	52	44	36	28	20	12	04
62	54	46	38	30	22	14	06	64	56	48	40	32	24	16	08
57	49	41	33	25	17	09	01	59	51	43	35	27	19	11	03
61	53	45	37	29	21	13	05	63	55	47	39	31	23	15	07

Fig 3: Example of permutation table of straight P-box where n is 64

P-Boxes..

- Example
  - Design an 8 × 8 permutation table for a straight P-box that moves the two middle bits (bits 4 and 5) in the input word to the two ends (bits 1 and 8) in the output words. Relative positions of other bits should not be changed.

P-Boxes..

- Example
  - Design an 8 × 8 permutation table for a straight P-box that moves the two middle bits (bits 4 and 5) in the input word to the two ends (bits 1 and 8) in the output words. Relative positions of other bits should not be changed.
- Solution
  - We need a straight P-box with the table [4 1 2 3 6 7 8 5].

P-Boxes..

- Compression P-Boxes
  - A compression P-box is a P-box with n inputs and m outputs where $m < n$
  - Some inputs are blocked and do not reach output
  - Normally are keyless with table
    - showing rule for transposing bits
    - This table has m entries with some missing values in output
  - Used when we need to permute and decrease number of bits for next stage at the same time

P-Boxes..

- Compression P-Boxes
  - Example of a 32 x 24 permutation table
  - Note that 7, 8, 9, 15, 26, 24 and 25 are blocked

01	02	03	21	22	26	27	28	29	13	14	17
18	19	20	04	05	06	10	11	12	30	31	32

Fig 4: Example of a 32 x 24 permutation table

P-Boxes..

- Expansion P-Boxes
  - An expansion P-box is a P-box with n inputs and m outputs where $m > n$
  - Some inputs are connected to more than one input
  - Normally are keyless with table
    - showing rule for transposing bits
    - This table has m entries with $m - n$ entries are repeated
  - Used when we need to permute and increase number of bits for next stage at the same time.

P-Boxes..

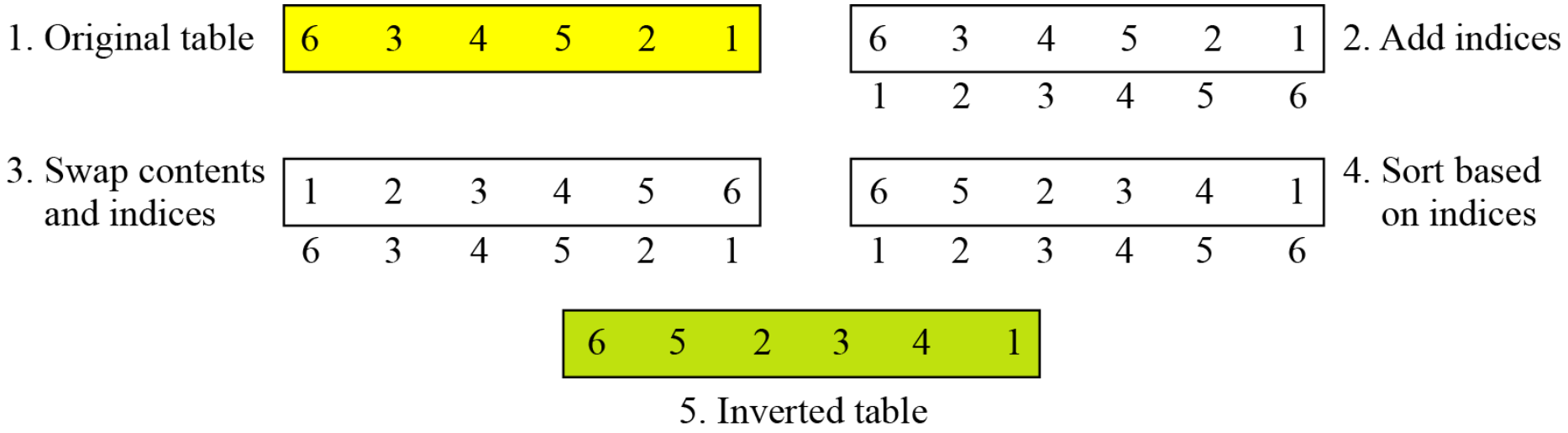
- Expansion P-Boxes
  - Example of a 12 X 16 P-Box
  - Note that inputs 1, 3, 9, and 12 are mapped to two outputs

01	09	10	11	12	01	02	03	03	04	05	06	07	08	09	12

Fig 5: Example of a 12 X 16 permutation table

P-Boxes..

- P-Box invertibility
  - What can you say about this???
  - A straight P-box is invertible, but compression and expansion P-boxes are not.
- Inverting a permutation table represented as a one-dimensional table.



P-Boxes..

- Compression and expansion P-boxes are non-invertible

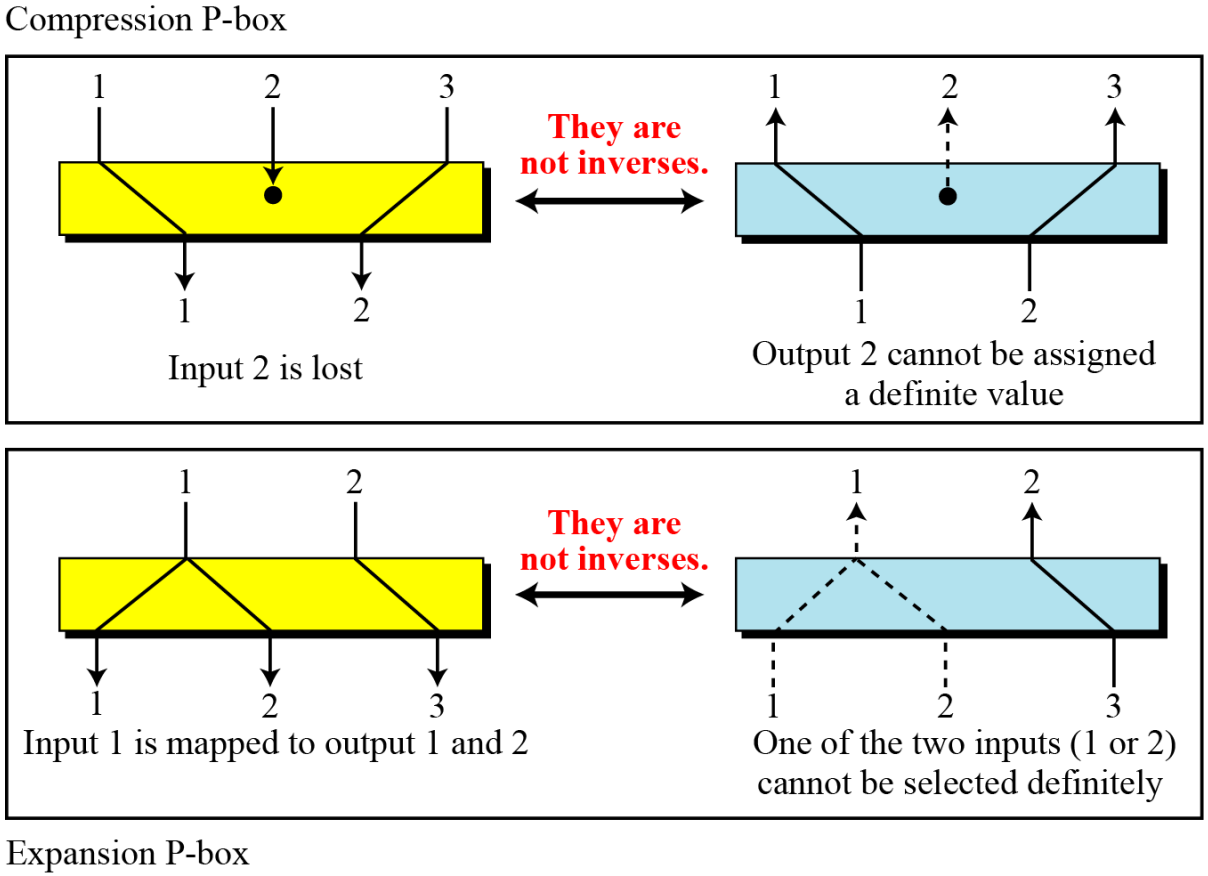


Fig 6: Compression and expansion P-boxes as non-invertible components

S-Boxes..

- An S-box (substitution box) can be thought of as a miniature substitution cipher.
  - An S-box is an $m \times n$ substitution unit, where m and n are not necessarily the same.
  - S-box can be keyed or keyless however,
    - modern block ciphers normally use keyless S-boxes, where mapping from input to output is predetermined
  - S boxes can be linear or nonlinear.

S-Boxes..

- Linear S-Box: Output can be expressed in linear transformation of input bits
  - Example
    - In an S-box with three inputs and two outputs, we have,

$$y_1 = x_1 \oplus x_2 \oplus x_3 \qquad y_2 = x_1$$

    - The S-box is linear because $a_{1,1} = a_{1,2} = a_{1,3} = a_{2,1} = 1$ and $a_{2,2} = a_{2,3} = 0$. The relationship can be represented by matrices, as shown below:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

S-Boxes..

- Nonlinear S-Box
  - In an S-box with three inputs and two outputs, we have

$$y_1 = (x_1)^3 + x_2 \qquad y_2 = (x_1)^2 + x_1x_2 + x_3$$

  - where multiplication and addition is in GF(2).
  - The S-box is nonlinear because there is no linear relationship between the inputs and the outputs.
  - Non-linear S-box have ‘AND’ terms in their expression

S-Boxes..

- Example
  - The following table defines the input/output relationship for an S-box of size 3 × 2.
  - The leftmost bit of the input defines the row; the two rightmost bits of the input define the column.
  - The two output bits are values on the cross section of the selected row and column.

Leftmost bit

	00	01	10	11
0	00	10	01	11
1	10	00	11	01

Rightmost bits

Output bits

  - Based on the table, an input of 010 yields the output 01. An input of 101 yields the output of 00.

S-Boxes..

- S-Box Invertibility
  - An S-box may or may not be invertible.
  - In an invertible S-box, the number of input bits should be the same as the number of output bits.

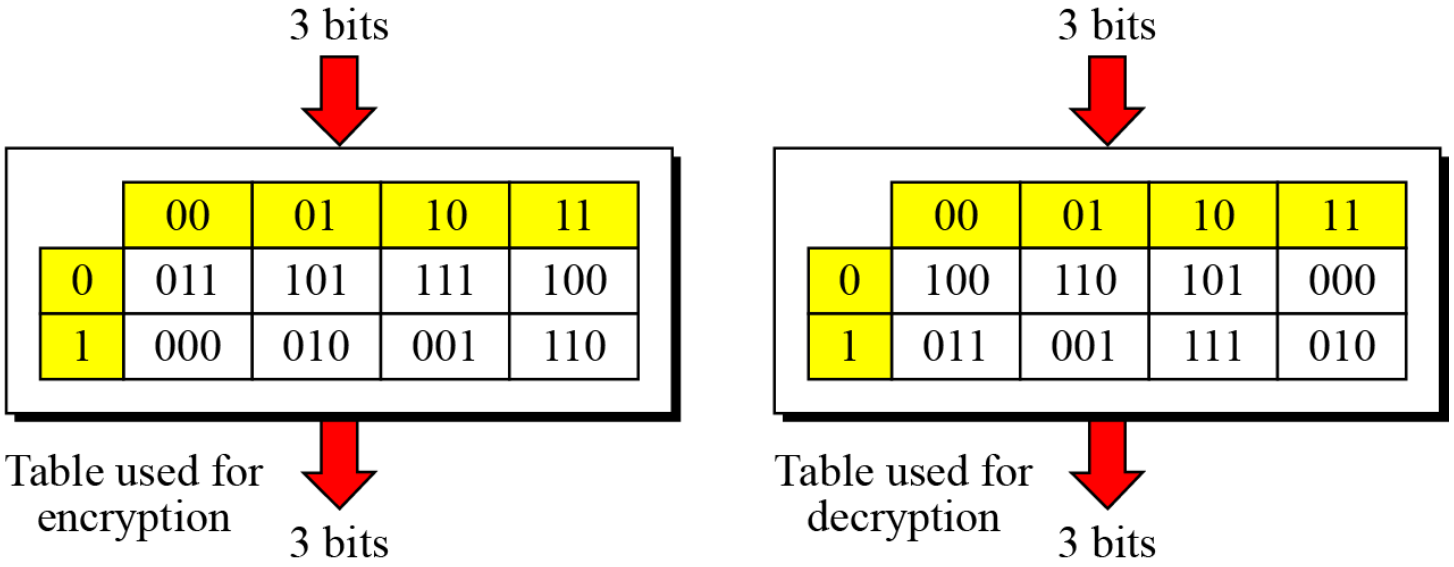


Fig 2: The S-box is invertible?

Components of a Modern Block Cipher...

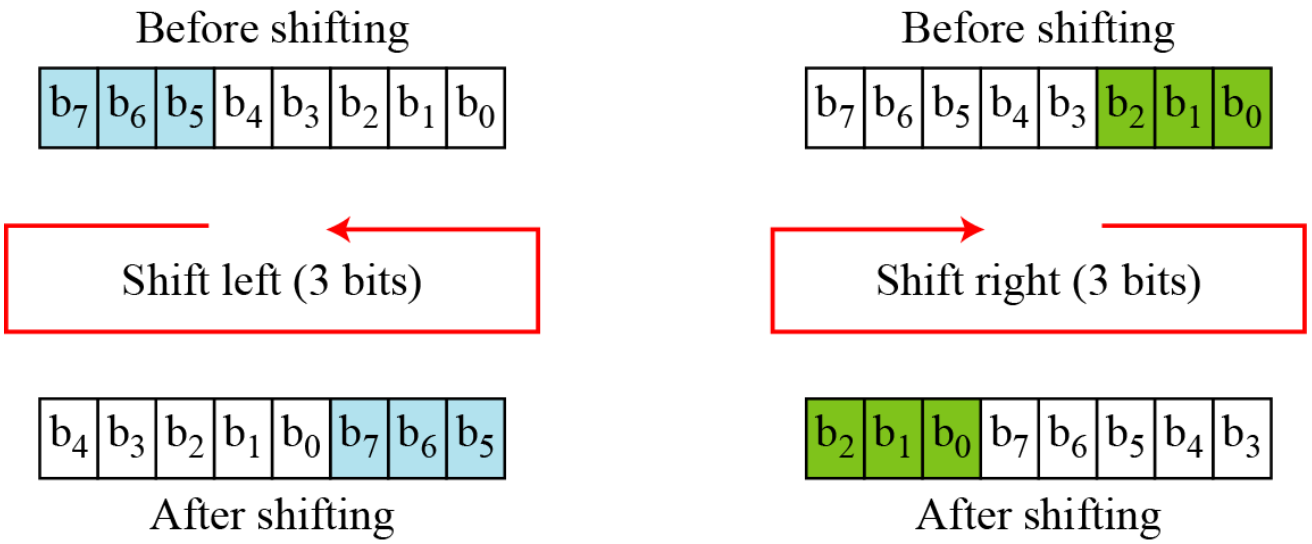
- Exclusive-Or...
  - An important component in most block ciphers is the exclusive-or operation.
  - Addition and subtraction operations in the GF(2^n) field are performed by a single operation called the exclusive-or (XOR).
  - The five properties of the exclusive-or operation in the GF(2^n) field makes this operation a very interesting component for use in a block cipher: closure, associativity, commutativity, existence of identity, and existence of inverse.

Components of a Modern Block Cipher...

- Inverse
  - The inverse of a component in a cipher makes sense if the component represents a unary operation (one input and one output).
    - For example, a keyless P-box or a keyless S-box can be made invertible because they have one input and one output.
    - An exclusive operation is a binary operation. The inverse of an exclusive-or operation can make sense only if one of the inputs is fixed (is the same in encryption and decryption).
      - For example, if one of the inputs is the key, which normally is the same in encryption and decryption, then an exclusive-or operation is self-invertible
        - $y = x \oplus k \quad \rightarrow \quad x = y \oplus k$

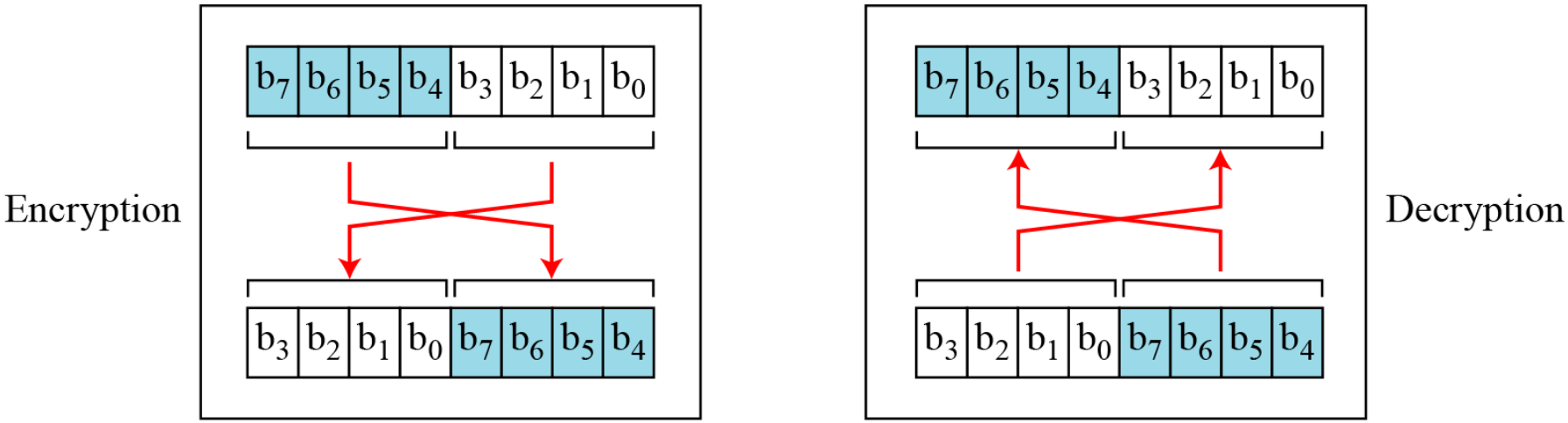
Components of a Modern Block Cipher…

- Circular Shift operation



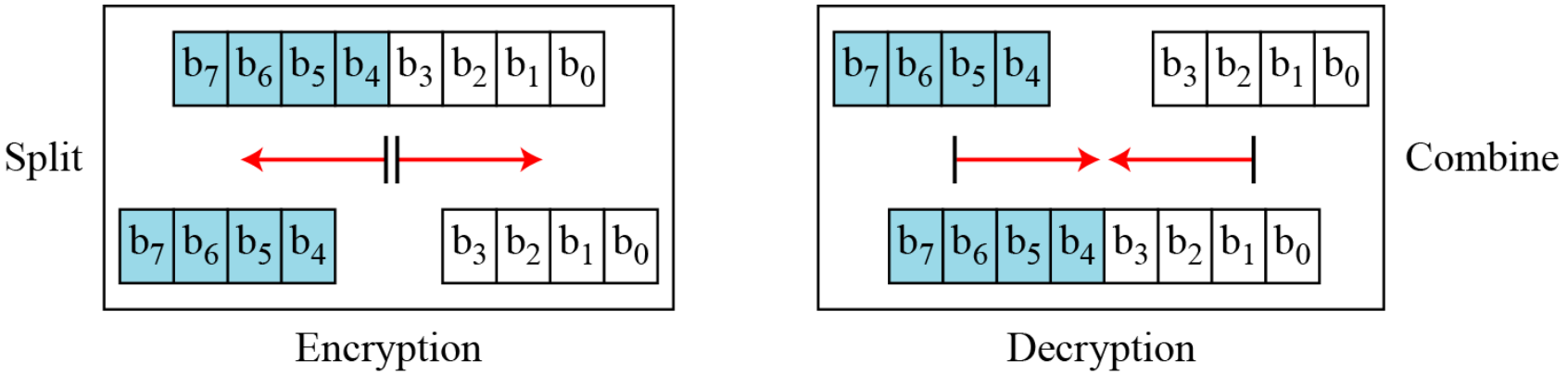
Components of a Modern Block Cipher…

- Swap
  - The swap operation is a special case of the circular shift operation where k = n/2.



Components of a Modern Block Cipher…

- Split and Combine
  - Two other operations found in some block ciphers are split and combine.
  - These operations are inverse of each other
    - if one is used for encryption, the other can be used for decryption



Product cipher

- Shannon introduced the concept of a product cipher.
- A product cipher is a complex cipher
  - combining substitution, permutation, and other components

Product cipher...

- Product Ciphers have two important properties:
  - Diffusion
  - Confusion
- Confusion
  - The idea of confusion is to hide
    - the relationship between the ciphertext and the key.
  - If single symbol of key is changed
    - several or all symbols of ciphertext will be changed
- Diffusion
  - The idea of diffusion is to hide
    - the relationship between the ciphertext and the plaintext.
  - If single symbol of plaintext is changed
    - several or all symbols of ciphertext will be changed

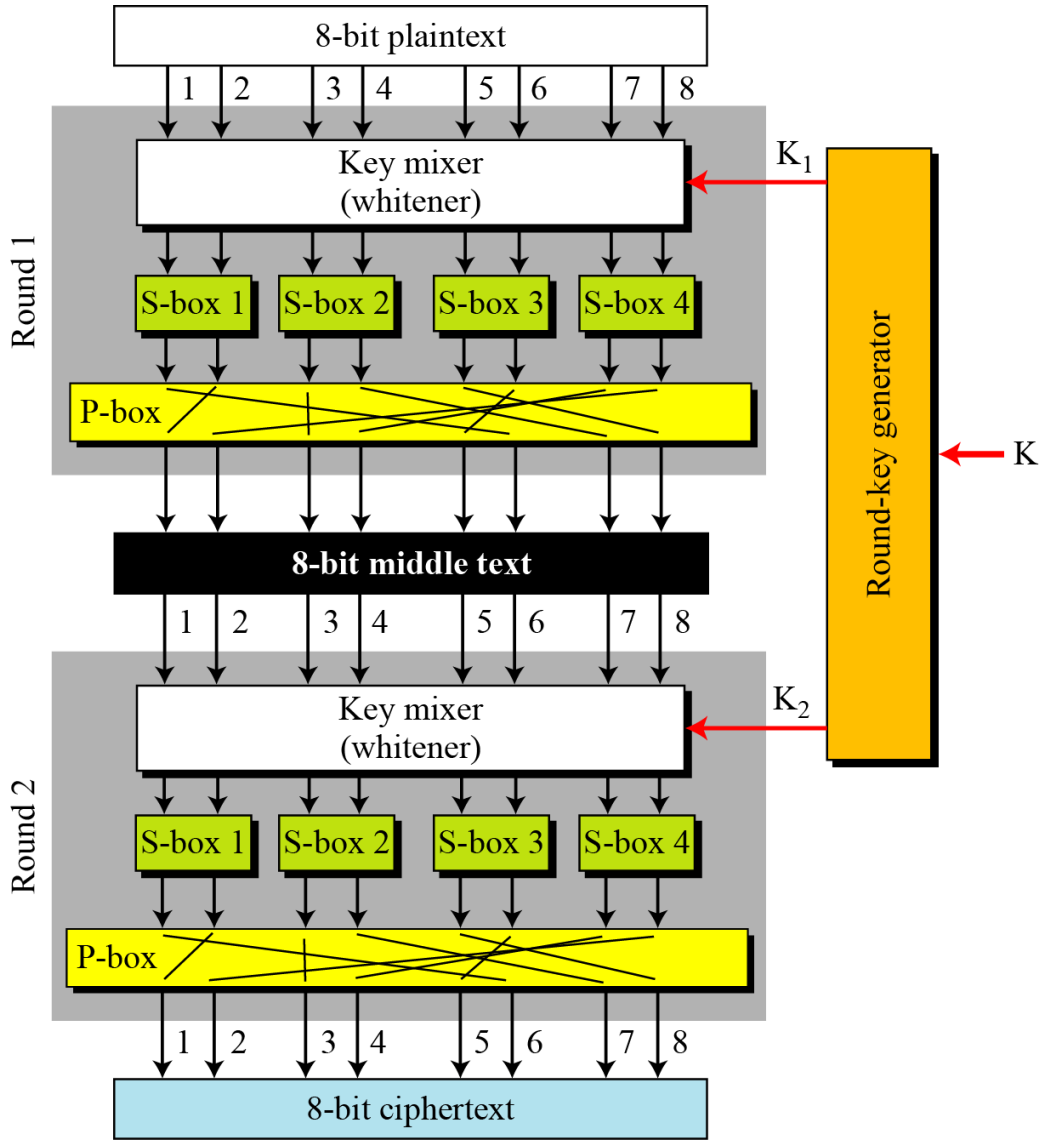
Product cipher...

- Diffusion and confusion can be achieved using iterated product ciphers where each iteration is a combination of S-boxes, P-boxes, and other components.
  - Each iteration is referred as round
- The block cipher uses **key schedule** or **key generator** that creates different keys for each round from the cipher key
- In an N-round cipher,
  - the plaintext is encrypted N times to create the ciphertext; the ciphertext is decrypted N times to create the plaintext.
  - Text created at the intermediate levels (between two rounds) as the middle text.

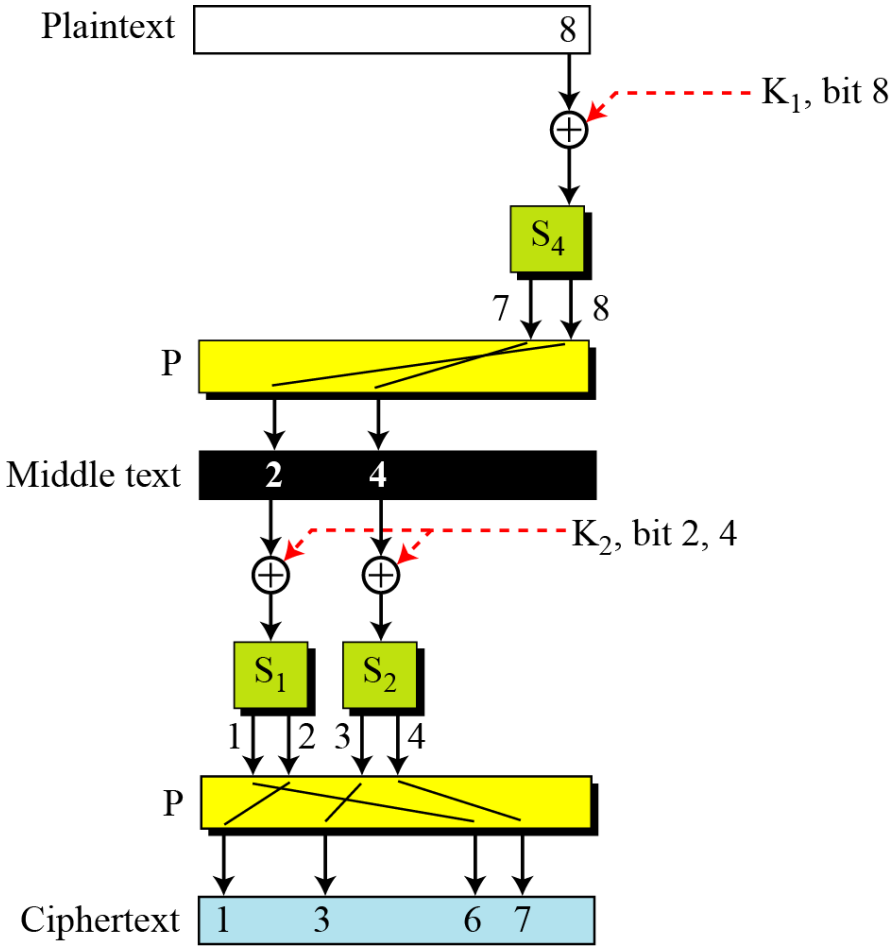
Product cipher Example

- An example of product ciphers that applies three transformations at each round:
  - The 8-bit text is mixed with the key to whiten the text (hide the bits using the key). This is normally done by exclusive-oring the 8-bit word with the 8-bit key.
  - The outputs of the whitener are organized into four 2-bit groups and are fed into four S-boxes. The values of bits are changed based on the structure of the S-boxes in this transformation.
  - The outputs of S-boxes are passed through a straight P-box to permute the bits so that in the next round each box receives different inputs.

Product cipher Example..



Product cipher Example..



Product Cipher..

- Practical Product ciphers use large blocks, more S-boxes and more number of rounds
  - to improve diffusion and confusion
- Increasing number of rounds and more S-boxes
  - May create better cipher in which ciphertext looks random n-bit word
    - Hides relationship between ciphertext and plaintext
- Increasing number of rounds
  - Increases number of round keys
    - Hides relationship between ciphertext and key

Two classes of product ciphers

- Modern block ciphers are all product ciphers, but they are divided into two classes.
  - Feistel ciphers
    - uses both invertible and noninvertible components.
  - Non-Feistel ciphers
    - use only invertible components

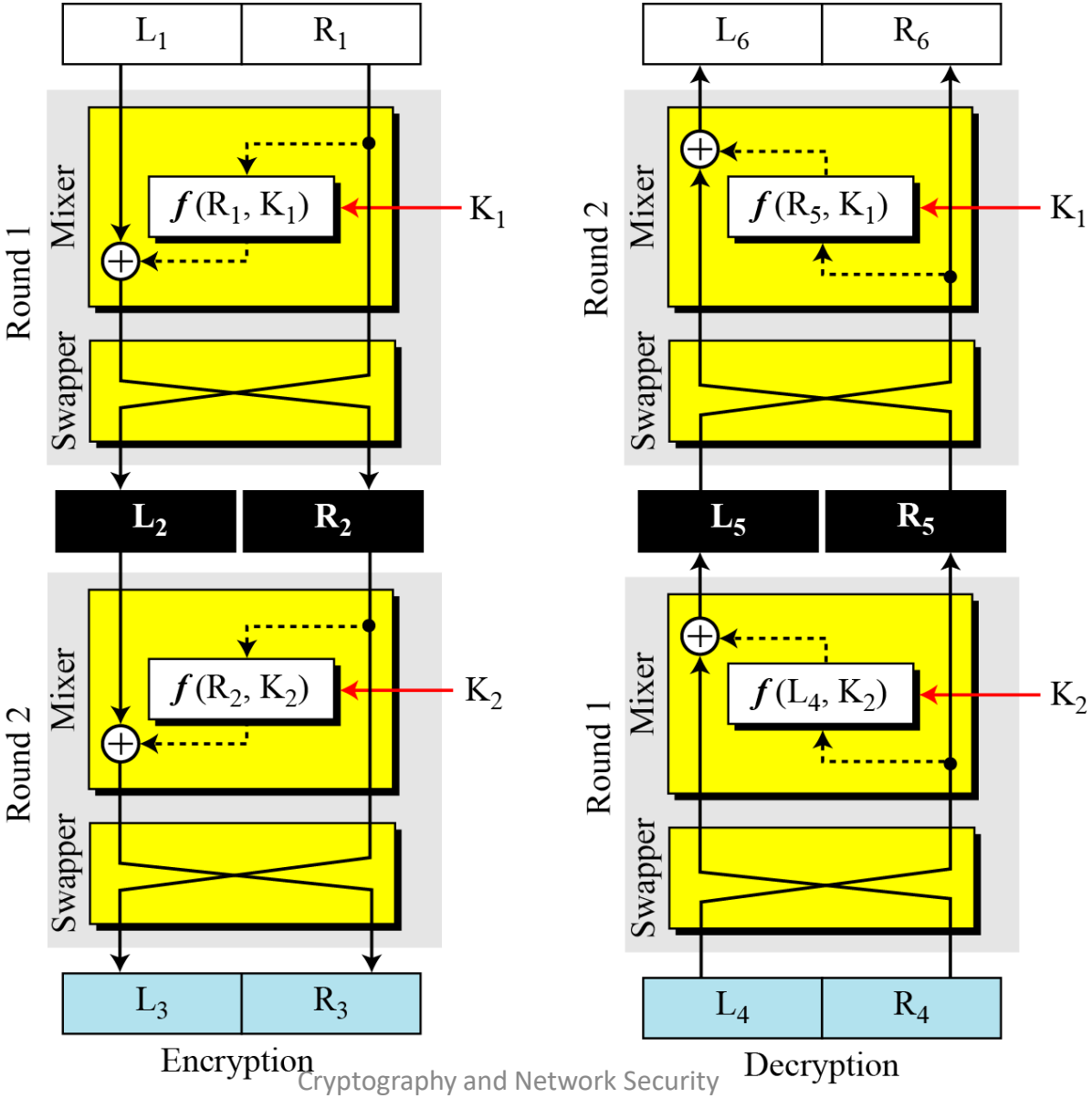
Feistel Cipher

- Feistel Ciphers
  - Feistel designed a very intelligent and interesting cipher that has been used for decades.
  - A Feistel cipher can have three types of components: self-invertible, invertible, and non-invertible.

How can encryption and decryption algorithms be inverses of each other if each has a noninvertible unit?

use exclusive-or operation

Final design of a Feistel Cipher...



Non-Feistel cipher

- Non-Feistel cipher
  - Use only invertible components
  - A component in the encryption cipher has the corresponding component in the decryption cipher.

Attacks on modern ciphers

- Attacks on traditional ciphers can also be used on modern block ciphers,
  - but today's block ciphers resist most of the attacks that are possible in classical ciphers